University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2019

Mark Sheet

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Question 1 [10 Marks]

a) Name two benefits of the Analyse-Develop-Test development cycle of Agile. [2]
Reduced risk [1], increased value as deliver benefits early [1], more flexibility/agility [1] Better cost management [1] (any two)

b) Why is continuous integration strongly encouraged in Agile? [2]
Ensures modules fit together [1] and product continues to work with all the changes [1]

c) How does test-driven development turn traditional development around, with respect to functional code and testing code? [2]
First write test code [1/2], then functional [1/2]. Traditional development is other way around [1]

d) By what name is the chain of tasks that determines the overall project duration known? [1]
Critical path [1] (accept only this)

e) In Scrum, what are the two meetings held after a Sprint? [1]
(Sprint) Review [1/2] and (Sprint) Retrospective [1/2] ('Sprint' not essential)

f) What are the differences between a vertical prototype and a horizontal prototype? [2]
Horizontal: wide range of functions [1/2] with little detail [1/2]. Vertical: lots of detail [1/2] For few functions [1/2]

Question 2 [10 Marks]

a) Explain your answers in each case below:

i. Is $O(2^n) \subseteq O(n!)$? [1]

ii. Is $O(\log_2 n) \subseteq O(1)$? [1]

iii. Is $\Theta(n) \subseteq \Theta(n!)$? [1]

Answer:

i) Yes. (½ mark). n! grows faster than 2^n (need to make an argument for growth rate) but Big Oh provides an upper bound (1/2 mark)

ii) No (1/2 mark). $\log_2 n$ grows faster than a constant (provide some kind of argument for this) (1/2 mark)

iii) No (1/2 mark). n! grows faster than n (make an argument for this) so they are in different efficiency classes. And the class is a subset of itself. (1/2 mark)

b) Consider the pseudocode below for Strassen's algorithm for matrix multiplication:

Fill A and B up with zeroes so as to get two n x n matrices where n is a power of 2. Then apply the algorithm below. [4]

```
Strassen(A, B):
if n == 1 return A00 × B00
else
  Split matrices into 8 n/2 X n/2 parts:
                    A00 , B00 , . . . , A11 , B11
  M3 = Strassen(A00, B01 – B11)
  M5 = Strassen(A00 + A01 , B11 )
  M2 = Strassen(A10 + A11 , B00 )
  M4 = Strassen(A11 , B10 – B00 )
  M1 = Strassen(A00 + A11 , B00 + B11 )
  M7 = Strassen(A01 – A11 , B10 + B11 )
  M6 = Strassen(A10 – A00 , B00 + B01 )
  C11 = M1 + M4 – M5 + M7
  C12 = M3 + M5
  C21 = M2 + M4
  C22 = M1 + M3 – M2 + M6
  return C
```

Set up a recurrence relation for the number of basic operations being performed by this algorithm and solve it using the Master Theorem.

Answer: $T(n) = 7T(n/2)+\Theta(n^2)$. (1 mark) a=7, b=2, d=2. (1 mark) So $7>2^2$. (1 mark) So T(n) is in $\Theta(n^{(\log_2 7)})=\Theta(n^{2.807})$. (1 mark). Don't need to get the 2.807 precise.

c) Give pseudocode for Floyd's algorithm to find the shortest path between all vertices, given a weighted connected graph. [2]

d) Explain how a weighted connected graph needs to be represented for Floyd's algorithm to work. [1]

Answer:

a.

```
floyd(W[1..n,1..n]
    for k=1 to n
      for i=1 to n
        for j=1 to n
          W[i,j] = min(W[i,j],W[i,k]+W[k,j])
```

b. Adjacency matrix where infinity represents no edge between two vertices, other values represent distance between vertices (so 0 between vertex x and itself).

Question 3 [10 Marks]

a) Can every NFA N be converted to an NFA N' with a single accept state that accepts the same language as N? If so, explain how to do so. If not, given an example. [2]

Yes it can. (1/2 mark) Informally, add a new state p (1/2 mark), add an epsilon transition from every accept state of N to p (1/2 mark) , and make p the only accept state. (1/2 mark)

b) Construct a context-free grammar that generates the following language: [4]

$L = \{w \mid w \in \{a,b,c\}^* \text{ and } w = w^r\}$. So every word in L is a palindrome made up of the letters a, b, and c.

$S \rightarrow X$ (1 mark)

$X \rightarrow aXa \mid bXb \mid cXc$ (1 mark)

$X \rightarrow a \mid b \mid c$ (1 mark)

$X \rightarrow \varepsilon$ (1 mark)

c) Prove that the context-free languages are closed under the * operation. [4]

Let G1 be a CFG that describes the language L1. Give each of the variables in G1 the subscript 1. Let S_1 be the start symbol for G1.

We construct a new CFG G from G1 that accept L1*. (1 mark)

Let S be a new variable (not occurring in G1). S is the start symbol for G. (1 mark) Add the following rules: $S \rightarrow \varepsilon \mid SS_1$. (1 mark for each rule)

Question 4 [10 Marks]

a) Cohesion and coupling are commonly referred to when talking about architectural design patterns. Explain what each of those two concepts mean, and what the goal of architectural design is in terms of those concepts. [3]

a. Answer: Cohesion is the degree to which communication takes place within the module [1]

b. Coupling is the degree to which communication takes place between modules [1]

c. Architectural design is about, Minimize coupling while maximizing cohesion [1]

b) What property defines a strict layered architecture? [1]

d. Answer: In a strict layered architecture, a layer only calls upon the services of the layer directly below it.

c) What is the lollipop notation used for in UML? [1]

e. Answer: To indicate type

d) Explain the concept of “separation of concerns” mean? [1]

f. Answer: separating a system into distinct sections, such that each section addresses a separate concern.

e) Which architectural design pattern is used for distributed systems over the internet such as email and the world wide web. [1]

Answer: Client Server

f) A compiler is a typical example of what kind of application architecture? [1]

Language processing systems

g) Briefly explain the Gang of Four Observer pattern and how it relates to the common MVC pattern. [2]

Answer: the object containing the data is separate from the objects which display the data [1]
Generalization of the MVC Pattern [1]